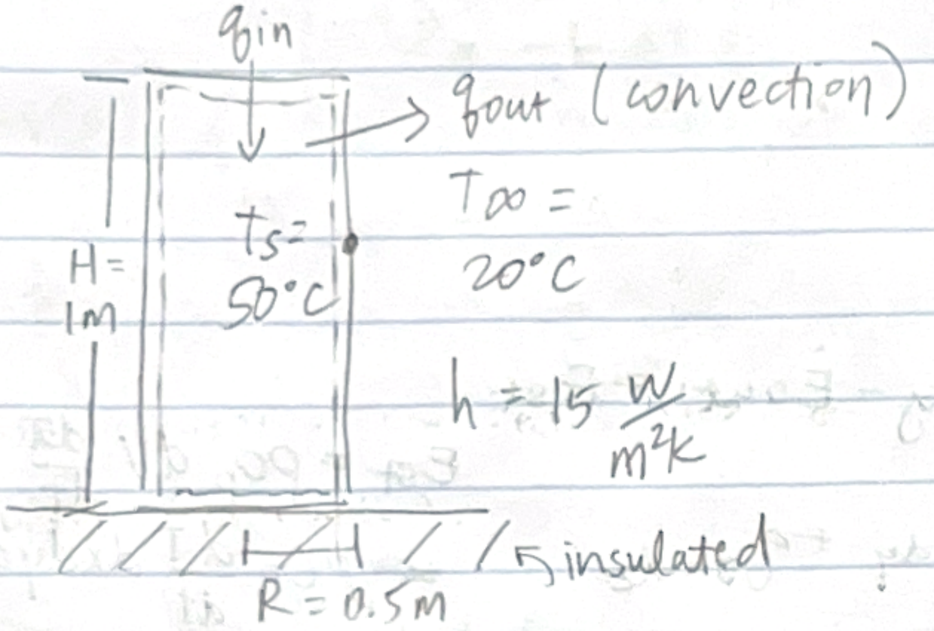
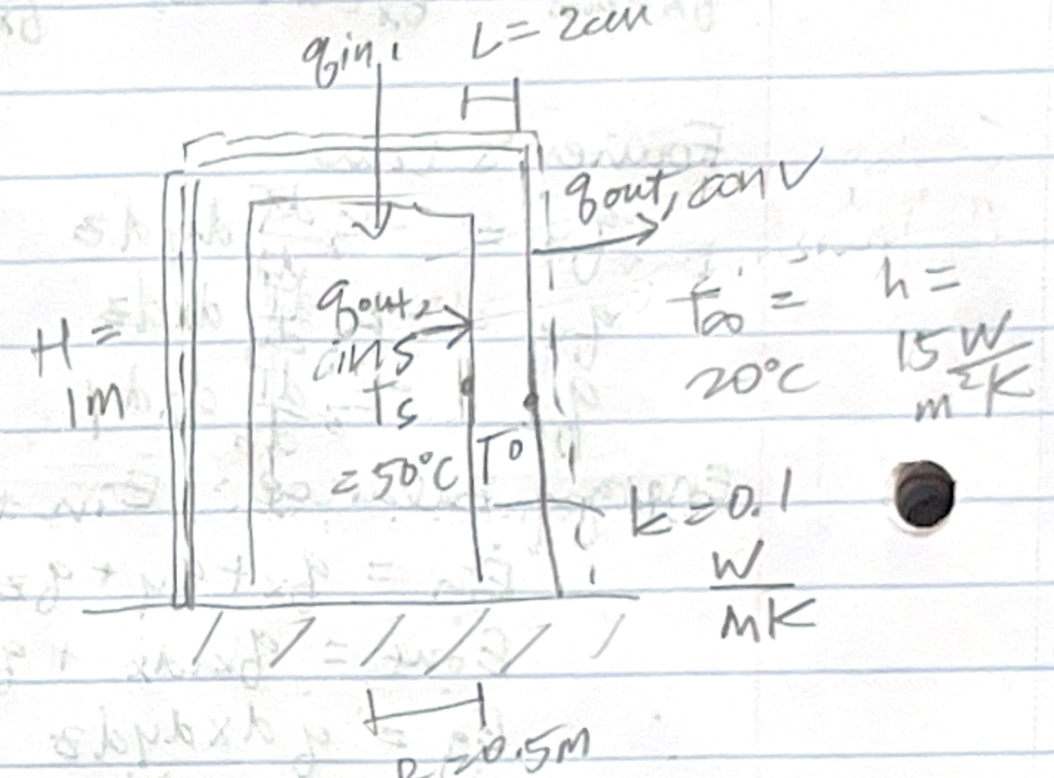
#1. Known: Cylindrical heater w/ $R=0.5m$, $H=1m$, $T_{in}=50°C=T_s$, $T_\infty=20°C$
$h=15\ W/m^2K$, insulation foam $k=0.1\ W/m\cdot K$

Find: a) q to maintain constant T @ 50°C, b) part a) but w/ $L=2cm$ foam
c) compare a) & b)

Schematic:





Assumptions: Steady state, constant air properties, negligible PE & KE change, thermally insulated bottom, negligible radiation

Analysis:

a) $q_{in} = q_{out}$

$= hA(T_\infty - T_s)$

$= 15 \cdot 3.927(20-50) = \boxed{1767.145\ W}$

Area of cylinder w/o bottom:

$\pi R^2 + 2\pi R \cdot H = \pi(0.5m)^2 + 2\pi \cdot 0.5m \cdot 1m$

b) $q_{in_1} = q''_{out} A$

$q''_{out} = h(T_0 - T_\infty) = \frac{k(T_0 - T_s)}{L}$

$q''_{out} = \frac{T_s - T_\infty}{\frac{L}{k} + \frac{1}{h}} = \frac{50-20}{\frac{0.02}{0.1} + \frac{1}{15}} = 112.5\ \frac{W}{m^2} \rightarrow q_{in} = 112.5 \cdot 3.927$

$= \boxed{441.788\ W}$

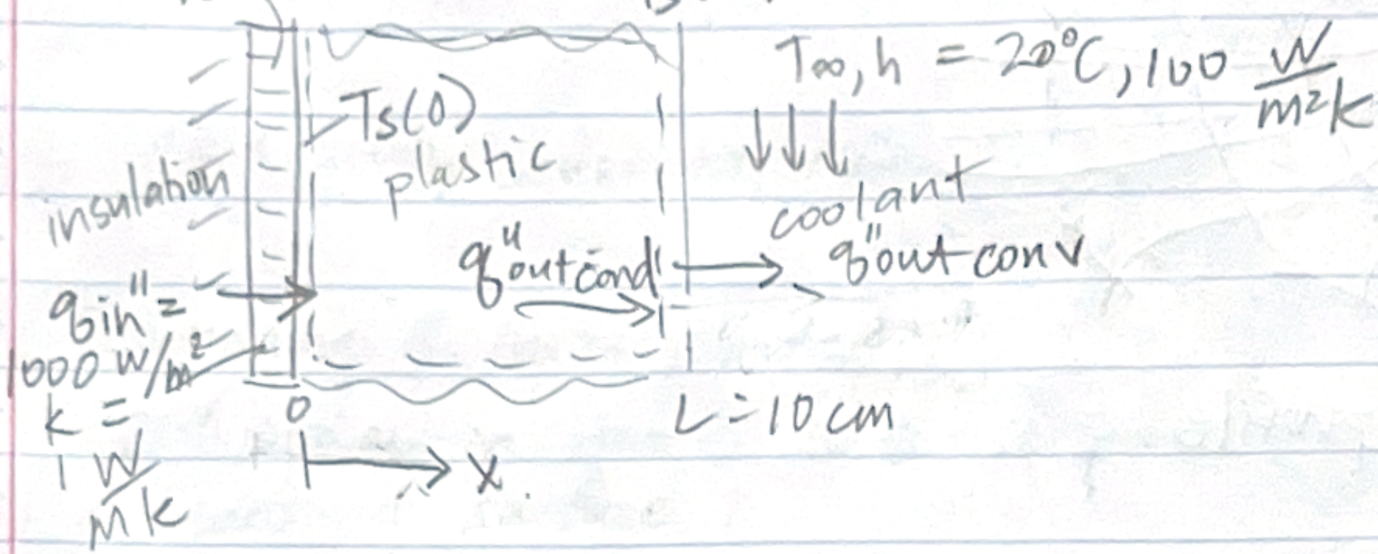
c) The insulation costs the water heater 4 times less power (1767 W vs 441 W).

cont.

#2. Known: Insulation is $L = 10\,cm$ thick, $k = 1 \frac{W}{mK}$. Coolant w/ $T_\infty = 20°$
$h = 100\ W/m^2K$

Find: a) when $q'' = 1000 \frac{W}{m^2}$, $T_s(x=0)$ & $(x=L)$, b) max heat flux if $T_s \leq$
c) 3 strategies to increase operation capacity w/o exceeding 200°C

Schematic:



Assumptions: Thermally insulated left side, neglect radiation, heater is infinitely thin, no KE/PE changes, steady state, constant air properties

Analysis:

a) $\dot{E}_{in} = \dot{E}_{out}$

$q''_{in} = q''_{out\,cond} = q''_{out\,conv}$

$1000 = \frac{k}{L}(T_s(0) - T_s(L)) = h(T_s(L) - T_\infty)$

$\rightarrow T_s(L) = \frac{1000}{h} + T_\infty = \frac{1000}{100} + 293 = \boxed{303\,K = T_s(L)}$

$\rightarrow T_s(0) = 1000 \cdot \frac{L}{k} + T_s(L) = 1000 \cdot \frac{0.1m}{1\,W/mK} + 303K = \boxed{403K = T_s(0)}$

b) $q'' = \frac{k}{L}(200°C - T_s(L)) = h(T_s(L) - 20°C)$

$T_s(L) = \frac{q''}{h} + 20°C$

$q'' = \frac{k}{L}(200°C - \frac{q''}{h} - 20°C) = 10(180 - \frac{q''}{h})$

$q'' + 10\frac{q''}{h} = 1800 \rightarrow q'' = \frac{1800}{1 + 10/h} = \frac{1800}{1 + 10/100} = \boxed{1636 \frac{W}{m^2}}$

c) 3 strategies include: decreasing width of plastic, choosing a coolant w/ higher h, choosing plastic with higher k

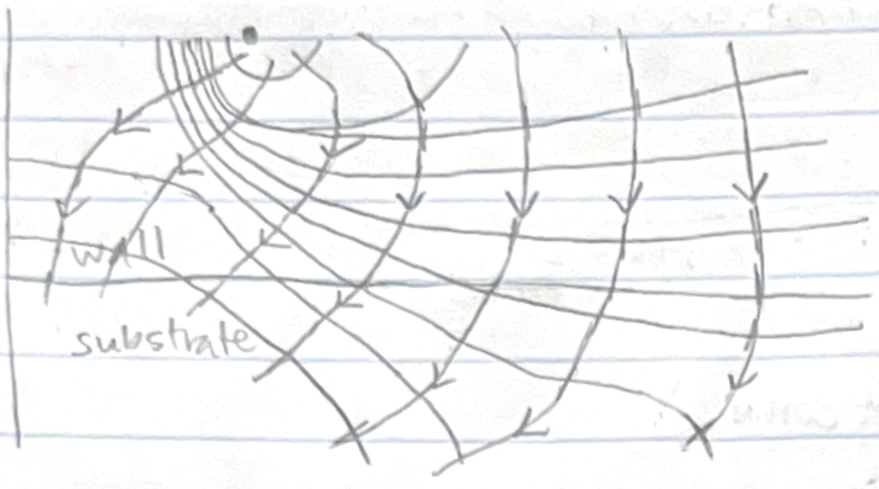
Cont.

#3. Known: heat flux is gradient of temperature profile

Find: heat flux vector in x-z plane, where laser has larger flux

Schematic:

a)



Assumptions: 2D profile

Analysis:

b) The left side has a larger heat flux because the isotherms are more closely stacked. Meaning, there's a steeper heat flux profile (gradient).

MAE 320 HW 2

Name: Allison Newman Date of submission:

Problem Set: #2 Due date: 2/10

Acknowledgement: I did not consult any classmates. I used course notes.